



University
of Glasgow

April/May Exam Session 2025
Duration: 1 hour 30 minutes
Timed exam fixed start time

DEGREES OF MSc, MSci, MEng, BEng, BSc, MA and MA (Social Sciences)

TEXT AS DATA FOR MSC/MSCI COMPSCI 5096/5106

(Answer all 3 questions)

This examination paper is an open book, online assessment and is worth a total of 60 marks.

1. **Question on TF-IDF and Set Similarity. (Total marks: 20)**

Consider the following vocabulary constructed from an unknown language. Inverse Document Frequency (IDF) values are provided and rounded to the nearest integer for ease of arithmetic.

Token ID	Token	IDF
0	A	2
1	B	4
2	C	1
3	D	3
4	E	6
5	F	4
6	G	1
7	H	3
8	I	3
9	J	5

- (a) Construct a dense TF-IDF bag-of-words representation for the following document. Use the IDF values provided above and linear TF scaling. Tokenize on whitespace.

DocX: G C G A H A G

[4]

- (b) Consider the following TF-IDF bag-of-words representation for DocY, which used the same procedure as in (a). Reconstruct the original text for DocY. Specify any ambiguities encountered during reconstruction.

DocY: [6, 8, 0, 6, 6, 0, 2, 3, 3, 10]

[4]

- (c) Compute the Overlap Coefficient and Jaccard Similarity between DocX and DocY. Show your work.

[4]

- (d) A colleague proposes a new set similarity measure:

$$\text{sim}(A, B) = \frac{|A - B|}{|A \cap B|}$$

Note that $A - B$ represents the set difference operation (in other words, the elements of set A that are not in set B). Compute $\text{sim}(\text{DocX}, \text{DocY})$. Show your work.

[2]

- (e) Describe **three** potential problems with the proposed similarity measure from (d).

[6]

2. Question on Text Classification. (Total marks: 20)

- (a) You are building a multi-class text classification system to categorize recipes for an existing recipe website using the text of the recipe. It has four classes: comfort food, easy-to-make, Italian and pasta.

To build your labelled dataset, you search the current website using the existing search functionality and the class names (e.g. searching for "comfort food"). You take the first five recipes retrieved for each class as your labelled data.

Discuss **four** problems that this approach may have.

[8]

- (b) You realise that samples with the "comfort food" label have been triplicated (repeated three times) by accident in the dataset. This error occurred before any data splitting occurred. What effect would this have on the training and evaluation? Give **four** points.

[4]

- (c) A separate classifier is trained and evaluated for identifying the sentiment of user comments on the reviews. Consider the statistics below. Identify **eight** anomalies in the dataset, the presented metrics and the overall training/evaluation procedure.

[8]

		Predicted	
		Pos	Neg
Actual	Pos	2	2
	Neg	8	3

Table 1: Confusion matrix for training set

Sample Count	15
Accuracy	0.5
Perplexity	19.2
Macro-Precision	0.2
Macro-Recall	0.5
Macro-F1	0.6

Table 2: Metrics for training set

		Predicted		
		Pos	Neg	Neutral
Actual	Pos	50	0	0
	Neg	0	10	0
	Neutral	0	0	5

Table 3: Confusion matrix for validation set

Sample Count	65
Accuracy	1.00
Perplexity	10.8
Macro-Precision	1.00
Macro-Recall	1.00
Macro-F1	1.00

Table 4: Metrics for validation set

3. **Question on Sentiment Analysis. (Total marks: 20)**

Consider a situation where you are working for a company, BeStep, that is looking to monitor its brand online. You've been hired to work on a project that tracks the sentiment (positive, negative, neutral, or mixed sentiment) towards the brand on social media.

(a) You are considering three possible approaches:

1. BERT-base, a pre-trained language model that is fine-tuned on your labelled sentiment data.
2. A logistic regression model that uses TF-IDF features and is fine-tuned on your labelled sentiment data.
3. LLAMA-70B, a Large Language Model that is prompted for sentiment.

Rank the three approaches in terms of their computational cost of deployment in this application, from least expensive to most expensive. Justify your answer.

[2]

(b) Describe a procedure for determining which of the three solutions is most effective in this setting. Clearly state any data you need to determine effectiveness in this setting.

[4]

(c) BeStep wants to expand internationally next year and, therefore, wants to ensure it is ready to monitor social media brand sentiment in languages other than English. Give one problem you would encounter when applying a logistic regression model in another language. Give a different limitation that you would encounter when applying a BERT-base classifier in another language. Suggest a solution that addresses both these problems.

[4]

(d) You consider a fourth approach, a BERT-based model that is fine-tuned on open benchmark data for sentiment classification. However, when you run it on your data, you find that it classifies every sample in your validation data that mentions the brand name as "positive" sentiment. You check the false positives (examples below) and confirm that their gold labels are correct.

- I disliked my experience with bestep.
- @Bestep I placed an order three weeks ago, and still haven't gotten it!
- I'm not happy with my purchase BeStep. I want a refund.

The problem is being caused by **sub-word tokenisation**. Describe the problem and provide one suggestion on how to address it.

[4]

(e) BeStep wants a more fine-grained understanding of the sentiment towards each of their 18 products. Describe an approach you would use to accomplish this goal and how it would fit into the existing sentiment classification system.

[6]